

MAXWELL KWESI MAC-OCLOO

East Lansing, MI, USA

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EDUCATION

Michigan State University

Ph.D. Biostatistics

Advisor: Professor David Todem

Aug 2023 – May 2028 (Expected)

East Lansing, MI

The University of Texas at El Paso

M.Sc. Statistics and Data Science

Advisor: Dr. Abhijit Mandal

Aug 2021 – May 2023

El Paso, TX

University of Energy and Natural Resources

B.Sc. Statistics (Valedictorian)

Advisor: Dr. Killian Asampana Asosega

Aug 2015 – Jul 2019

Sunyani, Ghana

RESEARCH INTERESTS

- Statistical Machine Learning
- Statistical Methodology
- Computational Statistics
- Causal Inference
- Non- and Semi-Parametric Models
- Survival Analysis / Time-to-Event Methods
- Latent Variable Modeling / SEM
- Longitudinal / Repeated-Measures Data Analysis

RESEARCH & TEACHING EXPERIENCE

Department of Epidemiology and Biostatistics, MSU

Graduate Research Assistant

Aug 2023 – Present

East Lansing, MI

- Developing nonparametric and semiparametric causal inference methods for continuous treatments, with a focus on dose-response estimation in observational health studies.
- Investigating penalized zero-inflated Poisson models for variable selection with time-varying covariates in longitudinal count data.
- Examining the role of social support and health behaviors in treatment adherence and patient survival, with emphasis on behavioral and social determinants of health equity.
- Collaborating with interdisciplinary teams to translate statistical findings into actionable insights for health research.
- Applied causal inference and missing data methodologies to clinical and epidemiological datasets using R and SAS.
- Developed and implemented statistical models (e.g., sequential mean models, regression, survival analysis) to evaluate treatment effects in observational studies using SAS.

Henry Ford Health, Public Health Sciences

Biostatistics Intern

May 2026 – Aug 2026

Detroit, MI

- Collaborated with investigators and interdisciplinary research teams to formulate research questions, select appropriate statistical methodologies, interpret findings, and translate statistical results into scientifically meaningful conclusions.
- Contributed to scientific manuscripts and research reports by developing statistical methods sections, interpreting analytical results, synthesizing relevant literature, and preparing results for peer-reviewed

dissemination.

- Designed and implemented statistical analyses for observational clinical and public health studies addressing treatment adherence, behavioral health, social support, substance use, and survival outcomes.

Department of Obstetrics, Gynecology and Reproductive Biology, MSU Jun 2025 – Apr 2026
Data Analyst East Lansing, MI

- Applied R and SAS to conduct statistical modeling, including causal inference, hazard modeling, trend analysis, and survival analysis, to investigate prognostic determinants of breast cancer mortality across racial groups.
- Analyzed racial disparities in breast cancer survival and evaluated temporal changes in mortality patterns across diagnosis periods.
- Conducted survival analyses and subgroup comparisons to assess the relationships between demographic, clinical, and tumor-related characteristics and breast cancer outcomes.
- Developed publication-quality visualizations, statistical reports, and analytical summaries to communicate findings to faculty investigators and research collaborators.

Lawrence Berkeley National Laboratory Jun 2023 – Aug 2023
Data Science Intern Berkeley, CA

- Turned large-scale high-performance computing (HPC) data into actionable insights on system efficiency, supporting real-time scientific data transfers between SLAC and NERSC.
- Built scalable Python pipelines to parse, integrate, and manage logs from HPC workflow managers and performance profilers.
- Designed and visualized performance metrics that revealed bottlenecks and usage patterns across big-data computing environments.
- Developed anomaly detection models to flag irregular system behavior and strengthen the reliability of HPC workflows.

The University of Texas at El Paso Aug 2021 – May 2023
Graduate Teaching Assistant El Paso, TX

- Developed and delivered lesson plans tailored to diverse learning needs, reinforcing core mathematical concepts and problem-solving skills.
- Administered, graded, and tracked weekly assignments and quizzes for 40 students to monitor academic progress and provide timely feedback.
- Tutored 55 students in Proportion and Algebra through structured and innovative instructional approaches.

Accra Institute of Technology April 2021 – Aug 2021
Academic Research and Teaching Assistant Accra, Ghana

- Supported the design, execution, and evaluation of research projects, conducting literature reviews, surveys, data collection, data integration, and statistical analysis in collaboration with faculty and research teams.
- Compiled, cleaned, coded, and analyzed research data; maintained research records and documented analytical procedures, results, and quality-control processes to support reproducible research workflows.
- Delivered undergraduate mathematics and statistics instruction through tutorials, seminars, laboratory sessions, and discussion sections, supporting students in the application of quantitative methods.
- Evaluated examinations, assignments, and student projects; provided timely feedback and supported faculty with assessment, grading, and development of instructional materials aligned with learning objectives.

- Contributed to curriculum development, academic seminars, student research projects, and academic administration, while maintaining current knowledge through professional development and engagement with emerging research methods.

Freelancer

Sep 2020 – March 2021

Independent Data Scientist

Accra, Ghana

- Provided independent data analysis and quantitative consulting services to individual clients, translating raw datasets and business questions into actionable statistical insights.
- Applied statistical methods and predictive modeling to support decision-making, forecasting, and prediction, including exploratory data analysis, trend analysis, and model-based inference.
- Cleaned, transformed, and analyzed client datasets using statistical and computational tools, developing summaries, visualizations, and analytical reports tailored to client objectives.
- Communicated analytical findings and model results to non-technical clients, translating quantitative evidence into practical recommendations to support planning and informed decision-making.

University of Energy and Natural Resources

Aug 2019 – Aug 2020

Teaching Assistant

Sunyani, Ghana

- Assisted courses in linear models, multivariate analysis, theory of interest, sampling theory, and statistical quality control.

PUBLICATIONS

- Ngozi Adaralegbe; **Maxwell K. Mac-Ocloo**; Minerva So; Mrudula Nair; Oluwale A Babatunde; Nosayaba Osazuwa-Peters; Amy M Williams; Eric Adjei Boakye. Patterns of Social Media Use and Suicidality among Middle School Students in the US. (*Under review*)
- Eric Adjei Boakye; **Maxwell K. Mac-Ocloo**; Nidhi Pawate; Mrudula Nair; Morgan Byrd; Nosayaba Osazuwa-Peters. Timing of HPV Vaccine Uptake Relative to CDC Guidelines in U.S. Youth. (*Preparing for resubmission*)
- Das, J.; Beyaztas, B.H.; **Mac-Ocloo, M.K.**; Majumdar, A.; Mandal, A. Testing Equality of Multiple Population Means under Contaminated Normal Model Using the Density Power Divergence. *Entropy* 2022, 24, 1189. <https://doi.org/10.3390/e24091189>
- Asosega, K.A., Opoku-Ameyaw, K., Otoo, D., **Mac-Ocloo, M.K.**, & Ayinzoya, R. (2020). Comparative Sensitivity Performance of the Discriminant Function and Logistic Regression under Different Training and Test Samples for Predicting Birth Outcomes. *Asian Journal of Probability and Statistics*, 6(3), 47–60. <https://doi.org/10.9734/ajpas/2020/v6i330163>

PRESENTATIONS

- **Internship Presentation** (Henry Ford Health, August 2026): Association between Social Support and Treatment Adherence in Head and Neck Cancer Patients.
- **Internship Poster Presentation** (Lawrence Berkeley National Laboratory, August 2023): Exploring Real-time Data Transfers from Stanford Linear Accelerator Center (SLAC) to National Energy Research Scientific Computing Center (NERSC).
- **Journal Club** (Michigan State University, January 2026): Variance Component Testing in Generalized Linear Models with Random Effects by Xihong Lin.

HONORS & AWARDS

- (2019) **Valedictorian**, University of Energy and Natural Resources graduating class (2019 Congregation).
- Awarded the **Best Student Award in the School of Sciences**, University of Energy and Natural Resources.
- Awarded the **Best Student Award in the Department of Mathematics and Statistics**, University of Energy and Natural Resources.
- (2026) Recipient of the **Graduate Office Fellowships**.
- (2025) Received a **Department of Epidemiology and Biostatistics Fellowship** for the 2025/2026 academic year.

SKILLS

Causal & Survival	Causal inference, mediation analysis, sequential mean models, survival and hazard modeling, trend analysis, missing data methods.
Statistical Modeling	Multivariable and mixed-effects regression, penalized regression and variable selection, robust and nonparametric inference, bootstrap inference, multiple testing.
Scientific Computing	R (R Studio, R Markdown), SAS, Python (PyTorch, NumPy, scikit-learn, SciPy), HPC.
Software & Workflow	Git/GitHub, reproducible research workflows, Streamlit, L ^A T _E X, Microsoft Office.
Languages	English (Fluent).

SERVICE & PROFESSIONAL AFFILIATIONS

Vice President	Biostatisticians and Epidemiologists at Michigan State (BEAMS), 2025 – 2027
Vice President	African Graduate Student Association (AGSA), MSU, 2026 – 2027
Student Member	American Statistical Association (ASA);
Student Member	Society for Industrial and Applied Mathematics (SIAM)
Student Member	Institute of Mathematical Statistics (IMS)